Predicting Customer Churn in the Telecommunication Industries Using Machine Learning Techniques

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Introduction

- Customer churn is a critical challenge that threatens profitability, sustainability, and competitiveness.
- Globally, churn rates in telecommunications range between 20–40%, leading to substantial revenue losses.
- Retaining customers is five to seven times cheaper than acquiring new ones (Kotler et al., 2021).
- In Sub-Saharan Africa, telecommunications access drives digital inclusion and economic participation (GSMA, 2022).

Problem Statement

- The telecommunications sector faces rising competition and evolving customer expectations Khan 2024, increasing the risk of churn.
- Existing churn prediction methods are largely reactive, Jahromi 2020 relying on static indicators and manual analysis.
- These traditional approaches are time-consuming, inefficient, and prone to human bias.
- They fail to capture behavioral and socio-economic changes, Khan 2024 leading to poor retention and revenue loss.

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Research Objectives

Primary Objective
To identify key churn predictors in telecommunication industry.

Specific Objectives
i. To describe the challenges and limitations of existing literature in churn prediction in the telecommunications industry.
ii. To develop a machine learning based churn prediction model using telecommunications data.
iii. To evaluate the performance of Machine Learning algorithms.
iv. To deploy Machine Learning models in an interactive dashboard.

Literature Review : Telecommunication Context

- Rapid technological change, market liberalization, and evolving customer needs have intensified competition in the telecom sector Chang2024. As services converge and switching costs drop, retaining customers has become a strategic priority. Firms must leverage data and digital engagement to differentiate and sustain loyalty.
- Global churn rates above 30% undermine profitability and raise acquisition costs Khan2024. Using machine learning models such as Random Forest and Gradient Boosting, Khan et al. (2024) achieved over 90% accuracy in predicting churn. Key drivers included poor network quality, pricing issues, and low engagement. Their findings stress the need for predictive analytics and targeted retention strategies to maintain competitiveness in a saturated market.

Literature Review : Theoretical Framework- CRM

- Customer experience theory emphasizes how digital touchpoints shape customer satisfaction, loyalty, and retention Smith2021. In competitive telecom markets, inconsistent service interactions and poor user experiences heighten switching behavior. Smith (2021) analyzed customer journey data using regression and satisfaction indices, revealing that seamless, responsive digital engagement reduces churn by up to 25%. The highlights experience management as a strategic factor in sustaining long-term loyalty.
- Customer Relationship Management frameworks use customer data to personalize offers and strengthen loyalty (Rahman2024).Rahman applied interpretable machine learning to telecom data, linking churn prediction with tailored promotions and loyalty programs. The study achieved high accuracy in identifying at-risk customers, showing that personalized engagement can raise retention by 20–30%. CRM-driven analytics thus shift churn management from reactive to proactive strategy.

Literature Review : Empirical evidence on churn prediction

- Nagarkar (2022) analyzed customer churn in Nepal's telecom sector using XGBoost on behavioral and recharge data. The study addressed the gap in regional churn analytics, where limited data-driven insights hinder retention strategies. Results showed that recharge frequency and inactivity duration were the strongest churn indicators, and model-guided interventions reduced customer loss by 17%. The work demonstrated the effectiveness of ensemble learning in identifying actionable behavioral drivers.
- Poudel and Sharma (2024) advanced churn prediction through explainer-aware models that integrate interpretability with predictive accuracy. Using feature attribution methods and cross-validation, their approach achieved AUC values above 0.90. The study emphasized transparency in AI-driven decisions, enabling telecoms to understand churn causes and design targeted retention programs. This highlights the growing shift toward explainable and trustworthy AI in customer analytics.

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Literature Review : Relevance and Contribution

- Mwaura (2021) examined the wider effects of telecom churn in Sub-Saharan Africa, linking persistent churn rates above 30% to reduced access to mobile financial services, digital education, and SME growth. The study used survey and secondary data to show that frequent switching weakens trust in service providers and limits digital inclusion. Findings highlight that churn is not only a business issue but also a developmental constraint affecting economic participation.
- Shahabikargar and Kiani (2025) explored advanced churn prediction using real-time external data integration. Their comparative analysis showed that hybrid models combining transactional, behavioral, and contextual data outperformed static models by 10–15% in accuracy and improved decision transparency. The research demonstrates how dynamic and explainable AI systems enhance both predictive power and managerial trust in customer retention strategies.

Research Gaps

- Link predictive insights to practical retention strategies and digital inclusion outcomes.
- Metric Bias: Studies focus on accuracy and AUC, with little attention to profit-based or ROI (Return on Investment) metrics like ARPU (Average Revenue Per User).
- Interpretability: Many ML models offer limited transparency, reducing business stakeholders understanding and trust in the results.

Methodology: CRISP-DM Approach

- The study applies the CRISP-DM framework to guide a structured and scalable churn prediction process.
- Uses secondary data from customer churn dataset containing real records from a leading Bulgarian telecommunication operator.
- Datasets cover demographics, contracts, billing,customer calling behavior and service usage for telecommunication customers.
- Enables cross-market comparison and validation of model performance.
- Data use aligns with global ethical and governance standards.

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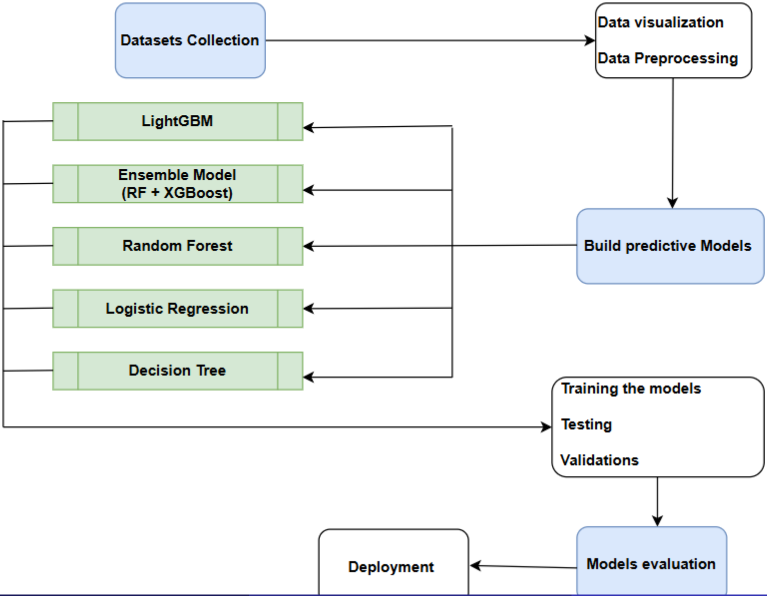
Methodology: Workflow



Methodology:Data Understanding and Cleaning

- A preliminary Exploratory Data Analysis (EDA) will summarize churn versus non-churn patterns using statistics, cross-tabs, and visualizations.
- Data cleaning will enhance quality by imputing missing values, correcting errors, and standardizing categorical/numeric variables.
- Outliers will be detected through statistical thresholds and visual diagnostics, then treated through capping, transformation, or removal.
- This preprocessing ensures a robust dataset for reliable predictive modeling.

- **Problem framing:** Customer churn is a binary classification task:

$$y_i = \begin{cases} 1, & \text{if CHURN} = 1 \text{ (customer churned)} \\ 0, & \text{if CHURN} = 0 \text{ (active customer)} \end{cases}$$

- **Comparative models:** Logistic Regression, Decision Tree, Random Forest, LightGBM, Ensemble models, and Logistic Regression with SMOTE to manage class imbalance.
- **Logistic Regression formulation:**

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \sum_{j=1}^{n} \beta_j x_j)}}$$

  where predictors include subscriber activity and revenue attributes.
- **Target interpretation:** $y_i = 1$ indicates customers lost within the observation period; $y_i = 0$ indicates retention.

Methodology:Machine Learning Modelling

- **Derived indicators:**

$$\text{ARPU} = \frac{\text{TotalRevenue}}{\text{Total\_SUBs}}, \quad \text{ActiveRate} = \frac{\text{Active\_subscribers}}{\text{Total\_SUBs}}, \quad \text{Suspensic}$$

$$\text{RevenueMix} = \frac{\text{AvgMobileRevenue}}{\text{AvgMobileRevenue} + \text{AvgFIXRevenue}}$$

- **SMOTE resampling (balancing minority churn class):**

$$x_{\text{new}} = x_i + \lambda(x_{nn} - x_i), \quad \lambda \in [0, 1]$$

- **Feature scaling:**

$$z = \frac{x - \mu}{\sigma}$$

  applied to numeric fields (`ARPU`, `ActiveRate`, `RevenueMix`, etc.) to stabilize learning.

- **Encoding:** Categorical variables (`CRM_PID_Value_Segment`, `EffectiveSegment`, `KA_name`) encoded via One-Hot Encoding; redundant or highly correlated features removed to enhance model interpretability and efficiency.

Methodology: Evaluation and Deployment

- **Data split:** Models trained using an 80/20 stratified split to preserve the churn class balance:

$$\text{Train:Test} = 0.8 : 0.2, \quad P(y=1)_{\text{train}} \approx P(y=1)_{\text{test}}$$

- **Evaluation metrics:** Performance assessed using:

$$\text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad \text{ROC} = \int_0^1 TPR(FPR)\, d(\text{FPR})$$

  Recall prioritized to minimize false negatives customers incorrectly predicted as non-churners.

- **Model deployment:** The best-performing model will be deployed through a RESTful API integrated with the CRM system, enabling churn probability outputs

$$\hat{P}(\text{CHURN} = 1|X_i)$$

  supporting targeted retention campaigns at scale.

Expected Outcomes

Technical Outcomes

- High model performance across key metrics (Recall, Precision, F1-score, ROC-AUC).
- Scalable architecture for seamless CRM integration and real-time predictions.

Operational Outcomes

- Proactive retention strategies to reduce revenue loss and improve loyalty.
- Optimized resource allocation targeting high-risk, high-value customers.

Broader Impact

- Provides a replicable framework for African telecommunication markets.
- Strengthens data-driven decision-making and digital inclusion.

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Conclusion

- Proposes a machine learning based churn prediction framework for telecoms.
- Tests multiple models (LR, RF, DT, LightGBM, Ensembles) for best performance.
- Integrates the final model with CRM systems for real-time insights.
- Aims to reduce churn, boost retention, and guide data-driven decisions.
- Offers a scalable, transferable framework for African telecommunication markets.

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